# Ansh Chandrakar

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## ABOUT ME

MSc Physics student at IIT Hyderabad with a strong foundation in computational physics. Passionate about exploring astrophysics and high-energy phenomena through numerical modeling and simulation. Experienced in utilizing Python to bridge theoretical physics with computational reality.

## EDUCATION

**Master of Science in Physics** 2025 - Present
**Indian Institute of Technology, Hyderabad, Telangana**

- *Core Courses*: Electrodynamics, Classical Mechanics, Quantum Mechanics - 1, Statistical Mechanics, Mathematical Physics - 1

**Bachelor of Science (Honours) in Physics** 2022 - 2025
**Birla Institute of Technology, Mesra, Ranchi, Jharkhand**

- *CGPA*: 8.6/10
- *Core Courses*: Mechanics, Mathematical Physics - 1, 2 and 3, Waves and Optics, Electricity and Magnetism, Thermal Physics, Digital Systems and Applications, Elements of Modern Physics, Analog Systems and Applications, Quantum Mechanics and Applications, Solid State Physics, Electromagnetic Theory, Statistical Mechanics
- *Elective Courses*: Computational Physics Skills, Radiation Safety, Nuclear and Particle Physics, Classical Dynamics, Astronomy and Astrophysics

**High School** 2017 - 2021
**Sri Sankara Vidyalaya, Bhilai, Chhattisgarh**

- *12th*: 91.8%
- *10th*: 92.4%

## RESEARCH PROJECTS

**MSc Winter Project: Non-Reciprocal Ising Model with Stochastic Resetting**

- Studied non-equilibrium statistical mechanics of Ising systems with asymmetric (non-reciprocal) interactions that violate detailed balance.
- Analyzed stochastic spin-flip dynamics and the emergence of non-equilibrium steady states with probability currents.
- Investigated the effects of stochastic resetting on system dynamics and steady states, highlighting the interplay between non-reciprocity and resetting.
- Explored broader implications for non-equilibrium phase behavior, with extensions proposed to Potts and XY-type models.

**Pedagogical Study on Quantum Entanglement (Manuscript in Preparation)**

- Preparing a pedagogical article focused on understanding quantum entanglement through the lens of coupled quantum harmonic oscillators, aiming to make the concepts accessible to a wider physics audience.
- Developing clear demonstrations and illustrations using Python (QuTiP) simulations to highlight key ideas such as Von Neumann entropy, entanglement dynamics, and decoherence.

- Targeting submission to *Resonance* journal to share findings and teaching-oriented insights.

**Bachelor's Dissertation: Entanglement in Coupled Quantum Oscillators Using QuTiP**

- Explored entanglement in coupled quantum harmonic oscillators using the Quantum Toolbox in Python (QuTiP) as part of the Bachelor's dissertation project.
- Investigated coupling mechanisms, unitary evolution, and computed entanglement measures such as Von Neumann entropy and logarithmic negativity across various systems.
- Extended the study to hybrid systems coupling quantum oscillators with coherent states, analyzing entanglement dynamics and decoherence effects.

**Academic Coursework & Modeling**

- **Galactic Cosmic Ray Propagation:** Initiated numerical modeling framework to simulate high-energy particle diffusion through galactic magnetic fields and the interstellar medium.
- **Fiber Optic Gyroscope (FOG) Modeling:** Collaborating on theoretical and computational modeling of the Sagnac effect to simulate phase shifts introduced by rotational motion.

## SKILLS

**Programming & Computing:** Python (QuTiP, NumPy, SciPy), MATLAB, Fortran, Bash Scripting, Linux System Administration.

**Typesetting & Tools:** LaTeX, Git/GitHub.

**Languages:** English (Fluent), Hindi (Native).

## EXPERIENCE

**Winter School on Recent Frontiers in Physics Research (Online)** December 2025
**Indian Institute of Science (IISc), Bengaluru**

- Participated in advanced lectures covering astrophysics and cosmology, quantum condensed matter, soft and active matter, plasma physics, photonics, and quantum technologies.
- Took part in tutorial sessions involving computational demonstrations and numerical exercises to illustrate physical concepts.
- Engaged with faculty and researchers through interactive discussions, with particular interest in lectures related to astrophysics and cosmology.
- Certificate of Participation

**IIA Online Summer School** June 2025
**Indian Institute of Astrophysics, Bengaluru**

- Selected to participate in a summer school hosted by the Indian Institute of Astrophysics.
- Attended lectures and hands-on sessions on advanced topics including galactic structure, observational techniques, radio astronomy, cosmology, and solar physics.
- Engaged with leading scientists through interactive sessions to deepen conceptual and practical understanding of astrophysics.
- Certificate of Completion

**Society for Data Science (Member)** Feb 2023 – May 2025
**BIT, Mesra, Ranchi**

- **Upskilled** in data analysis and programming through intensive Python workshops.
- **Spearheaded** logistics and event coordination for society-wide technical initiatives.
- **Secured** event sponsorships by pitching society initiatives to external partners.

## CERTIFICATIONS & AWARDS

- **IIT JAM 2025**: Achieved All India Rank (AIR) 220 in the Joint Admission Test for Masters.
- BIT GP Birla Scholarship for Program Topper – Awarded twice (2 semesters) for outstanding academic performance at BIT Mesra